

Complex Numbers: Cartesian Form & Algebra

ANSWER KEY



Basic arithmetic with complex numbers

- 1 Given $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$, find $z_1 + z_2$ and $z_1 - z_2$.
 $z_1 + z_2 = (3 + 1) + (2 - 4)i = 4 - 2i$. $z_1 - z_2 = (3 - 1) + (2 - (-4))i = 2 + 6i$.
- 2 Find $z_1 z_2$ for $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$.
 $z_1 z_2 = (3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2 = 3 - 10i - 8(-1) = 3 - 10i + 8 = 11 - 10i$.

The complex conjugate and division

- 3 Find the complex conjugate of $z = 5 - 3i$, and compute $z\bar{z}$.
 $\bar{z} = 5 + 3i$. $z\bar{z} = (5 - 3i)(5 + 3i) = 25 - 9i^2 = 25 + 9 = 34$ (a real number, equal to $|z|^2$).
- 4 Simplify $\frac{2+i}{3-2i}$ by multiplying by the conjugate of the denominator.
 $\frac{2+i}{3-2i} \cdot \frac{3+2i}{3+2i} = \frac{(2+i)(3+2i)}{9+4} = \frac{6+4i+3i+2i^2}{13} = \frac{6+7i-2}{13} = \frac{4+7i}{13} = \frac{4}{13} + \frac{7}{13}i$.

Modulus and the Argand diagram

- 5 Find the modulus $|z|$ of $z = 6 - 8i$.
 $|z| = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.
- 6 The graph shows several complex numbers plotted as points on the Argand diagram (real part on the horizontal axis, imaginary part on the vertical axis).
- a Read off the complex number plotted closest to the point (3, 2) in Cartesian (real, imaginary) form. The point (3, 2) corresponds to the complex number $3 + 2i$.
- b Find the modulus of the complex number plotted at (-3, -3). The point (-3, -3) corresponds to $-3 - 3i$, with modulus $|-3 - 3i| = \sqrt{(-3)^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}$.

Solving equations with complex roots

- 7 Solve $x^2 + 4x + 13 = 0$, giving your answer in the form $a + bi$.
Using the quadratic formula: $x = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i$.
- 8 One root of a quadratic equation with real coefficients is $2 - 5i$. State the other root, and find the quadratic equation.
Since coefficients are real, complex roots occur in conjugate pairs, so the other root is $2 + 5i$. Sum of roots = 4; product of roots = $(2 - 5i)(2 + 5i) = 4 + 25 = 29$. Equation: $x^2 - 4x + 29 = 0$.

Equating real and imaginary parts

- 9 Find real numbers x and y such that $(x + yi)(2 - i) = 8 + i$.
Expand: $(x + yi)(2 - i) = 2x - xi + 2yi - yi^2 = 2x + y + (2y - x)i$. Equate real and imaginary parts:
 $2x + y = 8$ and $2y - x = 1$. From the first, $y = 8 - 2x$. Substitute: $2(8 - 2x) - x = 1$, so
 $16 - 4x - x = 1$, $16 - 5x = 1$, $x = 3$. Then $y = 8 - 6 = 2$. So $x = 3$, $y = 2$.
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Powers of complex numbers using algebra

- 10 Find z^2 and z^3 for $z = 1 + i$, using direct algebraic expansion (without polar form).
 $z^2 = (1 + i)^2 = 1 + 2i + i^2 = 1 + 2i - 1 = 2i$. $z^3 = z^2 \cdot z = 2i(1 + i) = 2i + 2i^2 = 2i - 2 = -2 + 2i$.
- 11 Simplify i^{23} , using the fact that powers of i cycle with period 4 ($i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$).
 $23 = 4(5) + 3$, so $i^{23} = (i^4)^5 \cdot i^3 = 1^5 \cdot (-i) = -i$.
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The geometry of complex conjugates and reflections

- 12 Explain, in terms of the Argand diagram, what geometric transformation taking the conjugate of a complex number represents, and state the conjugate of $z = -4 + 7i$.
Taking the complex conjugate reflects the point representing z across the real axis (the horizontal axis) on the Argand diagram — the real part stays the same while the sign of the imaginary part flips. The conjugate of $z = -4 + 7i$ is $\bar{z} = -4 - 7i$.
- 13 If $z = a + bi$, express $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ in terms of z and $\bar{z}$.
Since $z + \bar{z} = (a + bi) + (a - bi) = 2a$, we get $\operatorname{Re}(z) = a = \frac{z + \bar{z}}{2}$. Since $z - \bar{z} = (a + bi) - (a - bi) = 2bi$, we get $\operatorname{Im}(z) = b = \frac{z - \bar{z}}{2i}$.
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Solving simultaneous equations with complex numbers

- 14 Solve the simultaneous equations $z_1 + z_2 = 5 + 3i$ and $z_1 - z_2 = 1 + 7i$ for complex numbers z_1 and z_2 .
Adding the two equations: $2z_1 = 6 + 10i$, so $z_1 = 3 + 5i$. Subtracting: $2z_2 = 4 - 4i$, so $z_2 = 2 - 2i$.
- 15 Given that $z = 2 + 3i$ is a root of $z^3 - 5z^2 + 17z - 13 = 0$ (a cubic with real coefficients), state the second complex root immediately (without further calculation), then find the third (real) root.
Since the cubic has real coefficients, complex roots occur in conjugate pairs, so $z = 2 - 3i$ is also a root immediately. The two complex roots multiply to $(2 + 3i)(2 - 3i) = 4 + 9 = 13$ and sum to 4, so together they correspond to the quadratic factor $z^2 - 4z + 13$. Dividing the cubic by this factor:
 $z^3 - 5z^2 + 17z - 13 \div (z^2 - 4z + 13) = z - 1$. So the third root is $z = 1$.

Loci and regions in the complex plane

- 16** Describe, geometrically, the locus of points z satisfying $|z - 3 - 4i| = 5$, and state its center and radius.
The equation $|z - (3 + 4i)| = 5$ describes all points z whose distance from the fixed point $3 + 4i$ equals 5. This is a circle centered at $(3, 4)$ with radius 5.
- 17** Find the two square roots of $z = 5 + 12i$ algebraically, by writing $\sqrt{5 + 12i} = a + bi$ and solving the resulting simultaneous equations $a^2 - b^2 = 5$ and $2ab = 12$.
From $2ab = 12$, $b = \frac{6}{a}$. Substitute into $a^2 - b^2 = 5$: $a^2 - \frac{36}{a^2} = 5$, so $a^4 - 5a^2 - 36 = 0$. Let $u = a^2$: $u^2 - 5u - 36 = 0$, factoring $(u - 9)(u + 4) = 0$, so $u = 9$ (rejecting the negative root since $a^2 \geq 0$). Then $a = \pm 3$, and $b = \frac{6}{a} = \pm 2$ (matching sign). Roots: $z = 3 + 2i$ or $z = -3 - 2i$.

Complex number identities and proof

- 18** Prove that $|z_1 z_2| = |z_1| |z_2|$ for any complex numbers $z_1 = a + bi$ and $z_2 = c + di$, by expanding both sides directly.
 $z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$. So
 $|z_1 z_2|^2 = (ac - bd)^2 + (ad + bc)^2 = a^2 c^2 - 2abcd + b^2 d^2 + a^2 d^2 + 2abcd + b^2 c^2 = a^2 c^2 + b^2 d^2 + a^2 d^2 + b^2 c^2$.
Also $|z_1|^2 |z_2|^2 = (a^2 + b^2)(c^2 + d^2)$, matching exactly. Since both sides are non-negative, taking square roots gives $|z_1 z_2| = |z_1| |z_2|$.
- 19** Given $z = x + yi$ satisfies $z^2 = \bar{z}$, find all complex solutions z .
 $z^2 = (x + yi)^2 = (x^2 - y^2) + 2xyi$, and $\bar{z} = x - yi$. Equating real and imaginary parts: $x^2 - y^2 = x$ and $2xy = -y$. From the second equation, either $y = 0$ or $x = -\frac{1}{2}$. Case $y = 0$: $x^2 = x$, so $x = 0$ or $x = 1$, giving $z = 0$ or $z = 1$. Case $x = -\frac{1}{2}$: $\frac{1}{4} - y^2 = -\frac{1}{2}$, so $y^2 = \frac{3}{4}$, giving $y = \pm \frac{\sqrt{3}}{2}$, so $z = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$. All solutions: $z = 0, 1, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i$.